Noc-Dashboard

Virtual Machine Setup Guide

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NOC-DASHBOARD

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1 Overview

This guide walks through getting Noc-Dashboard running on a fresh Windows VM, from an extracted deploy zip to a working, auto-starting dashboard.

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1.1 What This Covers

- Creating and activating a Python virtual environment for the app.
- Installing dependencies, including common failure points and how to fix them.
- Starting the dashboard and confirming it loads, from the command line or by double-clicking a desktop icon.
- What each helper script (start_dashboard.bat, create_shortcuts.bat, configure.bat) actually does internally.
- A settings window for card visibility and sync/alert intervals.
- Making the dashboard start automatically after a VM reboot.

NOTE

Run each command one at a time — don't chain steps together with &. Every command below that runs Python explicitly uses the venv's own interpreter, .venv\Scripts\python.exe (or .venv/Scripts/python.exe in Git Bash) — never a bare python — so it always runs with this project's installed dependencies, not whatever Python happens to be first on the VM's PATH.

2 Installation

Steps 1 through 5 get the application and its dependencies onto the VM. Run them in order.

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2.1 Unzip

Extract the deploy zip anywhere on the VM (e.g. E:\Noc-Dashboard).

2.2 Create the Virtual Environment

Command
python -m venv .venv

This one command is the only step in this entire guide that uses a bare python — it has to, since .venv doesn't exist yet for .venv\Scripts\python.exe to point at. It creates a self-contained Python install under .venv\, with its own python.exe, separate from whatever Python is already on the VM.

NOTE

If this fails with a path error, check that python on PATH is a real Python install, not a Windows Store alias, and that the account running it has write access to the folder.

2.3 Activate It (Optional)

Shell	Command
Command Prompt	.venv\Scripts\activate
Git Bash	source .venv/Scripts/activate (after: sed -i 's/\r$//' .venv/Scripts/activate)

NOTE

The generated activate script can have CRLF line endings that break source in Git Bash. Either fix it once with the sed command above, or skip activation entirely and call .venv\Scripts\python.exe directly by path for every command in this guide — which is what every command below already does, and what scripts\start_dashboard.bat and scripts\configure.bat both do internally too (Section 3).

2.4 Upgrade pip

Command
.venv\Scripts\python.exe -m pip install --upgrade pip

NOTE

The VM's default pip is often old enough (e.g. 21.1.3) that it fails to pick a prebuilt wheel for some packages and tries to compile from source instead — which then fails with 'Microsoft Visual C++ 14.0 or greater is required' if no compiler is installed. Upgrading pip first avoids this.

2.5 Install Dependencies

Command
.venv\Scripts\python.exe -m pip install -r requirements.txt

requirements.txt pins every package the dashboard needs — Flask and its extensions, pymssql and psycopg2-binary for the two databases, waitress for production serving, and two packages that are easy to overlook because their features aren't obvious from the main screen: openpyxl (every .xlsx export — Map, Reports, Partner Contact) and Pillow (the table-image attachments each Telegram alert sends). Installing from requirements.txt in one pass gets all of them — there is no separate install step for these two.

IMPORTANT — If a package fails with "Could not find a version that satisfies ...": the VM's package index doesn't have that exact pinned version (this happened with requests==2.34.2 and PySide6==6.11.1 — both too new for this VM's mirror). The error lists every version the index actually has; pick one, update requirements.txt (and pyproject.toml's matching >= line), then re-run the install.

IMPORTANT — If a package fails trying to compile (e.g. greenlet): first check whether a prebuilt wheel exists at all, without touching the version pin:

Command
.venv\Scripts\python.exe -m pip install --only-binary=:all: -r requirements.txt

- If that succeeds, the problem was pip's resolver, not a missing wheel — make sure step 2.4 (pip upgrade) actually ran, then retry the full install.
- If that fails too, pin the package to a specific version confirmed to have a wheel, or as a last resort install "Microsoft C++ Build Tools" from the URL in the error.

2.6 Verifying the Install

Before starting the dashboard for the first time, a quick sanity check confirms every package actually landed rather than silently failing partway through step 2.5:

Command
.venv\Scripts\python.exe -m pip show openpyxl pillow flask pymssql psycopg2-binary waitress

Each package prints a Name/Version/Summary block if installed, or "WARNING: Package(s) not found" on its own line if missing. openpyxl and Pillow specifically matter even though nothing on the dashboard's main screen depends on them directly — openpyxl silently breaks every .xlsx export button (Map, Reports, Partner Contact) if missing, and Pillow silently breaks the Telegram alerts' table-image attachments (they fall back to a plain-text list instead, per app/services/telegram/*.py).

3 Running the Dashboard

Once dependencies are installed, the dashboard can be started from the command line or a desktop icon, and configured through a settings window. This chapter also explains what each helper script actually does internally, not just how to run it.

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3.1 Start the Dashboard

Command
scripts\start_dashboard.bat

Run this from Command Prompt, or simply double-click start_dashboard.bat in File Explorer — both do exactly the same thing. Internally, the script:

- Switches to the project root folder (so it works no matter where it's launched from).
- Checks that .venv\Scripts\python.exe exists — if the virtual environment from Section 2.2 is missing, it prints the three setup commands needed and stops, rather than failing with a confusing error.
- Reads PORT from .env (5000 by default in .env.example; falls back to 8000 if .env has no PORT line at all).
- Opens http://localhost:<PORT>/ in the default browser automatically, 5 seconds after starting — enough time for Flask to finish booting first.
- Runs the dashboard using .venv\Scripts\python.exe run.py — the same venv Python used throughout this guide, never a bare python.

NOTE

The Command Prompt window this opens IS the running server — closing it stops the dashboard and every background sync job with it. Minimize it instead of closing it to keep the dashboard running in the background.

3.2 Desktop Shortcuts (Recommended)

For day-to-day use, a desktop icon is simpler than opening a terminal every time. Run once:

Command
scripts\create_shortcuts.bat

Figure 3-1 The actual Noc Dashboard icon (app/static/images/noc_dashboard.ico) used for both desktop shortcuts.

This adds two icons to the Desktop, both using the icon above:

Icon	Double-click to...
Noc Dashboard	Runs scripts\start_dashboard.bat — starts the server and opens the browser, exactly as in Section 3.1.
Dashboard Settings	Runs scripts\configure.bat — opens the settings window from Section 3.3, no terminal window involved.

No administrator rights are needed — the script uses PowerShell's WScript.Shell to write two .lnk shortcut files directly to the current user's Desktop, then refreshes the Windows icon cache (ie4uinit.exe -show) so the icon above shows correctly right away instead of a generic placeholder. Safe to re-run any time; it simply overwrites both shortcuts.

3.3 Configuration

Double-click the Dashboard Settings desktop icon, or run scripts\configure.bat directly, to open a settings window (configurator.py) for toggling which dashboard cards are shown and editing sync/alert intervals — it saves back to .env. Internally, configure.bat launches configurator.py with .venv\Scripts\pythonw.exe specifically (the windowless variant of the venv's Python) rather than python.exe, so the settings window opens on its own with no black console window behind it.

Setting	Takes effect
Card visibility (FEATURE_* flags)	Live, on the next page load — no restart needed
Sync intervals & alert settings	Needs the dashboard restarted — these control background threads that only read .env once at startup

NOTE

configurator.py only ever rewrites the specific settings shown in its window — secrets it doesn't manage (DB_PASSWORD, MSSQL_PASSWORD, SECRET_KEY, and the rest of .env) are left untouched. .env already has this deployment's real database credentials — no changes needed unless this VM should point at different databases.

4 Auto-Start on Boot

By default the dashboard only starts when someone double-clicks the Noc Dashboard icon. If the VM reboots and nobody logs in and starts it manually, the dashboard and all its background sync jobs stay down until someone does. Pick one option below to fix that.

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4.1 Task Scheduler (Recommended)

Starts the dashboard automatically at boot, even if nobody logs in. Needs admin rights.

- Open Task Scheduler (taskschd.msc), choose Create Task (not "Create Basic Task").
- General tab: name it Noc Dashboard; check "Run whether user is logged on or not" and "Run with highest privileges".
- Triggers tab -> New -> Begin the task: "At startup".
- Actions tab -> New -> Start a program: the full path to scripts\start_dashboard.bat (e.g. E:\Noc-Dashboard\scripts\start_dashboard.bat), with "Start in" set to the project root folder — required, otherwise the .bat can't find the venv/app.
- Save, entering the VM account's password when prompted.
- Test: right-click the task -> Run, and confirm the dashboard starts.

Equivalent one-line version (elevated Command Prompt)
schtasks /create /tn "Noc Dashboard" /tr "E:\Noc-Dashboard\scripts\start_dashboard.bat" /sc onstart /ru SYSTEM

4.2 Startup Folder Shortcut

Simpler, but only starts the dashboard when that specific Windows account logs in — not on an unattended reboot. No admin rights needed.

- Press Win + R, type shell:startup, press Enter — this opens the Startup folder for the current user.
- Copy the Noc Dashboard desktop shortcut (created in Section 3.2) into that folder.
- Done — it now starts automatically the next time this account logs in.